UVC REFERENCE

1. Hardware

When using EVB as a USB device, please disconnect JP84 and connect JP87.

Check whether R456 and R457 are soldered (the default is not), if not, mini-USB cannot be used, only USB type A can be used.

2. Software

2.1. Kernel Config

There are two ways to increase the uvc config on the basis of the existing config:

1. Modify the current default configuration directly (recommended)

   Such as the current EVB config: pioneer3_ssc020a_s01a_defconfig

   Add the following contents:

   ```
   CONFIG_MEDIA_SUPPORT=m
   CONFIG_MEDIA_CAMERA_SUPPORT=y
   CONFIG_MEDIA_CONTROLLER=y
   CONFIG_VIDEO_DEV=m
   CONFIG_VIDEO_V4L2=m
   CONFIG_VIDEOBUF2_CORE=m
   ```

```
CONFIG_VIDEOBUF2_MEMOPS=m
CONFIG_VIDEOBUF2_VMALLOC=m
CONFIG_MEDIA_SUBDRV_AUTOSELECT=y
CONFIG_USB_GADGET=m
CONFIG_USB_GADGET_VBUS_DRAW=2
CONFIG_USB_GADGET_STORAGE_NUM_BUFFERS=2
CONFIG_USB_GADGET_SSTAR_DEVICE=m
CONFIG_USB_AVOID_SHORT_PACKET_IN_BULK_OUT_WITH_DMA_FOR_ETHERNET=y

CONFIG_USB_LIBCOMPOSITE=m
CONFIG_SS_GADGET=m
CONFIG_USB_F_UVC=m
CONFIG_USB_G_WEBCAM=m
CONFIG_USB_WEBCAM_UVC=y
CONFIG_MULTI_STREAM_FUNC_NUM=1
```

See diff file [media/pioneer3_ssc020a_s01a_defconfig.diff] for details.

2. Use current default config

   a. Code

   ```
   cd kernel;./list_config.sh

   make pioneer3_ssc020a_s01a_defconfig
   ```

   b. Add: make menuconfig

      i. media frame config

      ```
      Device Drivers

      Multimedia support

      Device Drivers

      Multimedia support

      Cameras/video grabbers support

      Device Drivers

      Multimedia support

      Media Controller API
      ```

Output module: media.ko videodev.ko v4l2-common.ko

ii. usb Gadget frame config

```
Device Drivers

USB support

Device Drivers

USB support

USB Gadget Support
```

Output module: usb-common.ko udc-core.ko

iii. udc driver config: this is a HW IP related module, configured according to the specific situation.

```
Device Drivers

USB support

USB Gadget Support

USB Peripheral Controller

Sstar USB 2.0 Device Controller
```

Output module: udc-msb250x.ko

iv. gadget webcam

```
Device Drivers

USB support

USB Gadget Support

USB Gadget Drivers

Device Drivers

USB support

USB Gadget Support
```

```
USB Gadget Drivers

USB Webcam Gadget

Device Drivers

USB support

USB Gadget Support

USB Peripheral Controller

Sstar USB 2.0 Device Controller

Avoid short packet in bulk out with DMA for ethernet
```

Output module: libcomposite.ko videobuf2-core.ko videobuf2-v4l2.ko

videobuf2-memops.ko videobuf2-vmalloc.ko

usb_f_uvc.ko g_webcam.ko

2.2. Project Config

Because there is no UVC related configuration added, the following files need to be manually modified when compiling the uvc function.

1. If you use the second way in the kernel config, please modify the project/makefile and comment `$(MAKE)linux-kernel` in image to avoid overwriting the manual compilation when recompiling the kernel;
2. Add the following cotents in `project/kbuild/customize/4.9.84/p3/dispcam/kernel_mod_list`.

```
media.ko
videodev.ko
v4l2-common.ko
videobuf2-core.ko use_mi_buf=1
videobuf2-v4l2.ko
videobuf2-memops.ko
videobuf2-vmalloc.ko
```

```
udc-core.ko
libcomposite.ko
usb_f_uvc.ko
udc-msb250x.ko
g_webcam.ko streaming_maxpacket=3072 streaming_maxburst=13
uac_function_enable=0
```

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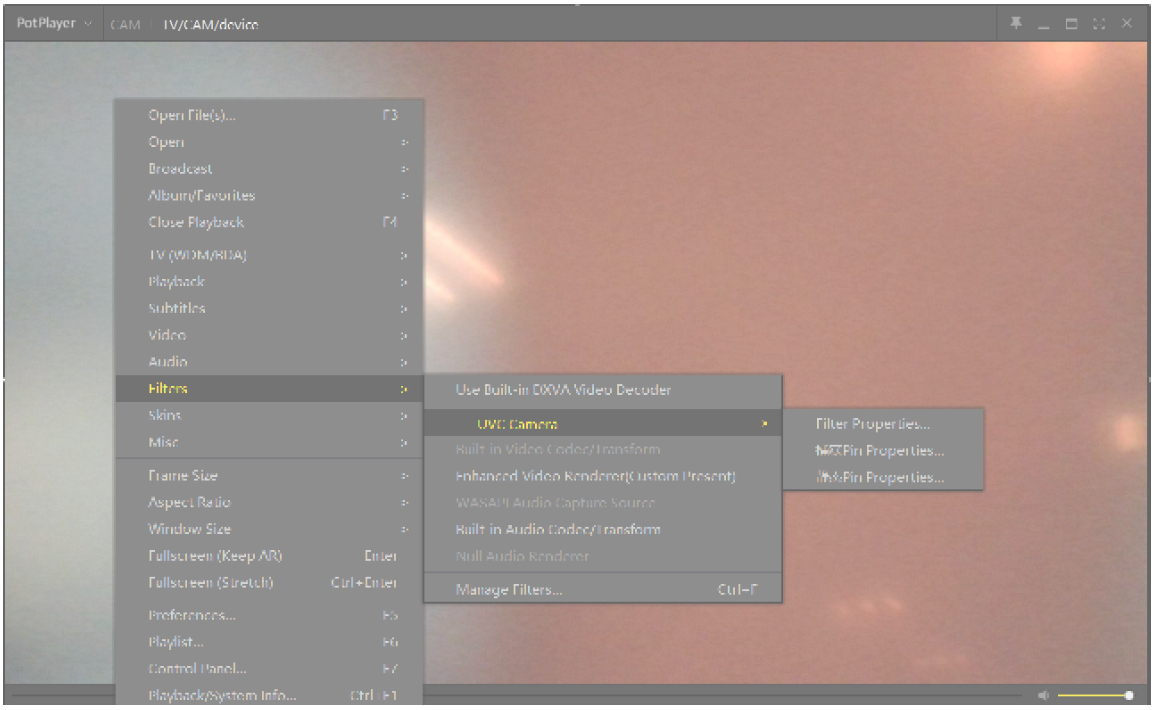
3. APP Test

Use mi_demo to test, enter `sdk/verify/mi\_demo/alderaan` and execute `make uvc`, after successful compilation, you can get `prog_uvc` in `sdk\verify\mi_demo\out\demo\app`, copy it to the board and run `./prog_uvc`. Without parameters, pad0 is used by default. Use `-a` to specify sensor pad (For example: `./prog_uvc -a1` specifies the image of pad 1), and use `prog_uvc-h` to print all supported parameters.

After prog_uvc is running, connect to PC by usb, and use potplayer (or other similar programs) to open the camera on PC.

Take potplayer as an example, use `ctrl+j` to open camera after running.

Right-click in the video preview area, you can modify the video stream parameters in the "Capture Properties" in the following path.


PotPlayer
CAM
TV/CAM/device
Open File(s)... F3
Open
Broadcast
Album/Favorites
Close Playback F4
TV (WDM/BDA)
Playback
Subtitles
Video
Audio
Filters
Skins
Misc
Frame Size
Aspect Ratio
Window Size
Fullscreen (Keep AR) Enter
Fullscreen (Stretch) Ctrl+Enter
Preferences... F5
Playlist... F6
Control Panel... F7
Playback/System Info... Ctrl+F1
Use Built-in DXVA Video Decoder
UVC Camera
Built-in Video Codec/Transform
Enhanced Video Renderer(Custom Present)
WASAPI Audio Capture Source
Built-in Audio Codec/Transform
Null Audio Renderer
Manage Filters... Ctrl+F
Filter Properties...
Pin Properties...
Pin Properties...